ANALYSTLAB AFRICA

Experience Lab Internship Programme

Week 4 Project Summary

HealthConnect Clinic Experience Lab — Data Analytics Track

Project	HealthConnect Clinic Experience Lab
Track	Data Analytics
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Week	Week 4 – Project Kickoff & Problem Understanding
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Week 4 Project Summary

1. The Specific Problem This Track Addresses

As the Data Analytics contributor to the HealthConnect project, my responsibility this week was to establish an evidence-based understanding of which factors are actually associated with appointment no-shows — rather than relying on assumptions from the business scenario. This foundation is intended to support both clinic decision-making and the Data Science track's later predictive modelling work.

2. Resources Used

- HealthConnect_Appointment_Data.csv — 5,000 appointment records across 18 variables.
- HealthConnect_Data_Dictionary.xlsx — variable definitions and structure reference.

3. Key Observations from Week 4

- The dataset is clean and internally consistent: no duplicate records, and no logical violations (e.g. no prior no-shows exceeding prior appointments, no bookings dated after their appointment).
- Only three variables show meaningful, evidence-backed separation between Attended and No-Show outcomes: previous_no_shows, booking_lead_days, and distance_to_clinic_km.
- Reminder delivery shows only a weak effect, which runs counter to the assumption embedded in the business scenario — an important, evidence-based finding in its own right.
- Appointment logistics (day, time, type) and patient demographics (age, gender) show no meaningful separation and were deprioritised for this analysis.
- Cancelled appointments are behaviourally distinct from No-Shows — a cancellation represents a patient successfully communicating non-attendance, while a no-show does not — and the two should never be merged into a single category.

4. Proposed Approach

The proposed approach is a segment-rate analysis built on the three strongest variables, supplemented by two composite KPIs: a Wasted Slot Rate (cut by reminder status) and a Cancellation-to-No-Show Ratio. Five business questions were defined, each mapped to a specific KPI, so that every question has a trackable, evidence-linked metric. Analysis will be computed in Python, with missing values in distance_to_clinic_km and waiting_time_minutes handled by documented exclusion rather than imputation, and visualisation deferred to Power BI or Tableau in a later phase.

5. Key Considerations That May Affect the Project

- Findings to date are correlational, not causal, and have not been tested for confounding between variables (e.g. lead time versus distance).
- The dataset is synthetic and includes future-dated records, confirming it is simulated rather than live clinic data — patterns hold within the dataset but are not validated against real-world behaviour.
- Some extreme values (e.g. a 100% no-show rate at previous_no_shows = 5) are based on very small sample sizes and should not be treated as stable patterns without further validation.
- The decision to keep Cancelled and No-Show as distinct outcome categories needs to be aligned with the Data Science track before they finalise their modelling target.

6. Proposed Focus for Week 5

- Calculate and visualise the five locked KPIs, rather than only identifying and justifying them.
- Test whether the three strongest variables hold up when examined in combination, to check for confounding.
- Coordinate with the Data Science track on the outcome-variable definition before their modelling work begins.

This summary complements the Week 4 Initial Analysis Document and Data Quality Assessment and should be read alongside them.